Hundred Days Of Code
Day--27

Question: Write a C program that displays the following menu and uses a switch statement to print the “Pyramid/Diamond pattern” selected by the user. Use nested loops without arrays. Each pattern may be implemented in a separate parameterless function with no return value. Note: For each activity, prepare the algorithm and test-case table. Then write, compile and execute the C program, and test it against the prepared test cases.

Solution:

. Aim

To write, compile and execute a menu-driven C program using switch statements and nested loops to display different pyramid and diamond patterns without using arrays.

2. Algorithm

1. Start the program.
2. Display a menu containing the following pattern choices:
    - Full Pyramid
    - Inverted Pyramid
    - Diamond Pattern
    - Palindromic Number Pyramid
    - Repeated Number Pyramid
    - Exit
3. Accept the user's choice.
4. Use a switch statement to select the corresponding pattern.
5. Input the number of rows (n).
6. Use nested loops to generate the selected pattern:
    - **Full Pyramid:** Print asterisks in increasing odd numbers in each row.
    - **Inverted Pyramid:** Print asterisks in decreasing odd numbers in each row.
    - **Diamond Pattern:** Print an increasing pyramid followed by a decreasing inverted pyramid.
    - **Palindromic Number Pyramid:** Print numbers in increasing order and then decreasing order in each row.
    - **Repeated Number Pyramid:** Print the current row number (2n − 1) times.

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7. Display the generated pattern.
8. Repeat the menu until the user selects Exit.
9. Stop.

C Program:

```
#include <stdio.h>


// Function to print Full Pyramid
void fullPyramid()
{
   int n, i, j;


   printf("Enter number of rows: ");
   scanf("%d", &n);


   for(i = 1; i <= n; i++)
   {
      for(j = 1; j <= 2*i-1; j++)
      {
         printf("* ");
      }
      printf("\n");
   }
}


// Function to print Inverted Pyramid
void invertedPyramid()
{
```

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```
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for(i = n; i >= 1; i--)
    {
        for(j = 1; j <= 2*i-1; j++)
        {
            printf("* ");
        }
        printf("\n");
    }
}

// Function to print Diamond Pattern
void diamondPattern()
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    // Upper half
    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= 2*i-1; j++)
```

```
    {
        printf("* ");
    }
    printf("\n");
  }

  // Lower half
  for(i = n-1; i >= 1; i--)
  {
    for(j = 1; j <= 2*i-1; j++)
    {
        printf("* ");
    }
    printf("\n");
  }
}

// Function to print Palindromic Number Pyramid
void palindromicPyramid()
{
  int n, i, j;

  printf("Enter number of rows: ");
  scanf("%d", &n);

  for(i = 1; i <= n; i++)
  {
```

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```
        // Increasing order
        for(j = 1; j <= i; j++)
        {
            printf("%d ", j);
        }

        // Decreasing order
        for(j = i-1; j >= 1; j--)
        {
            printf("%d ", j);
        }

        printf("\n");
    }
}

// Function to print Repeated Number Pyramid
void repeatedNumberPyramid()
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= 2*i-1; j++)
```

```
    {
        printf("%d ", i);
    }
    printf("\n");
  }
}

// Main Function
int main()
{
  int choice;

  do
  {
    printf("\n----- PYRAMID AND DIAMOND PATTERNS -----\n");
    printf("1. Full Pyramid\n");
    printf("2. Inverted Pyramid\n");
    printf("3. Diamond Pattern\n");
    printf("4. Palindromic Number Pyramid\n");
    printf("5. Repeated Number Pyramid\n");
    printf("6. Exit\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch(choice)
    {
```

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```
    case 1:
        fullPyramid();
        break;

    case 2:
        invertedPyramid();
        break;

    case 3:
        diamondPattern();
        break;

    case 4:
        palindromicPyramid();
        break;

    case 5:
        repeatedNumberPyramid();
        break;

    case 6:
        printf("Program exited successfully.\n");
        break;

    default:
        printf("Invalid choice! Please try again.\n");
}
```

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```
  } while(choice != 6);


  return 0;
}
```

Test Case Table:

Test Case	Menu Choice	Input (n)	Expected Output
1	1	4	Full pyramid with 4 rows
2	2	4	Inverted pyramid with 4 rows
3	3	4	Diamond with 7 rows
4	4	4	Palindromic number pyramid with 4 rows
5	5	4	Repeated number pyramid with 4 rows
6	1	1	Single row containing *
7	4	1	Single row containing 1
8	6	-	Program terminates
9	8	-	Displays invalid choice message

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Compilation and Execution:

